
Sarang Shastry

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EDUCATION

PhD in Computer Science (CSIS – Theoretical Computer Science)

July 25, 2026

University: BITS Pilani, K K Birla Goa Campus

Advisor: A. Baskar

Master of Computer Applications

Graduation: 2023

University: Gyan Ganga College of Technology

GPA: 8.58

Bachelor of Computer Applications

Graduation: 2021

University: Nachiketa College of Computer Science, Commerce, and Advanced Technology

Percentage: 61%

RESEARCH EXPERIENCE

Research Internship, BITS Pilani, K K Birla Goa Campus

May – July 2026

- Explored formal verification and interactive theorem proving using Lean 4.
- Studied mechanized reasoning for mathematical and computational properties in a proof assistant.
- Worked with foundational concepts connecting logic, formal methods, and machine-checked proofs.

Research Collaboration, Swansea University

Jan 2026 – Present

- Working on formalizing polynomial-time cryptographic adversaries using Rocq (Interactive Theorem Prover).
- Modeling adversarial behavior under polynomial-time constraints within a proof assistant.
- Studying foundational links between cryptography, computational complexity, and interactive theorem proving.

TECHNICAL SKILLS

- **Proof Assistants:** Coq (Rocq), Lean, NuSMV
- **Programming:** C++, Python, Java, OCaml, Haskell
- **Tools:** LaTeX, VS Code, Microsoft Office

RESEARCH INTERESTS

- Formal Verification
- Cryptography
- Mathematical Logic
- Interactive Theorem Proving

TEACHING EXPERIENCE

Faculty Member, Divysrijan IAS Academy

2024 – 2025

- Taught computer science and mentored students for exams
- Designed assignments and evaluated performance

Assistant Professor, Thevanai Ammal College

2023 – 2024

- Designed and delivered undergraduate CS courses like Java, Data Structures and Algorithms, Theory of Computation, and Discrete Mathematics.
- Used interactive teaching methods to improve learning

PROJECTS

Customer Churn Prediction using logistic regression

Python, scikit-learn

- Developed a machine learning model to predict customer churn using logistic regression on a telecom dataset.
- Performed data pre-processing, feature engineering, and model evaluation with scikit-learn, identifying key churn drivers like tenure and monthly charges.

Sales Analysis Dashboard

Tableau

- Designed an interactive Tableau dashboard for sales performance analysis, integrating data from multiple sources to visualize trends, regional sales, and top products.
- Enabled stakeholders to drill down into KPIs like revenue growth and customer segmentation.

SQL-based Data Analysis

Joins & Window Functions

- Conducted advanced SQL queries on e-commerce datasets using complex joins, window functions (e.g., ROW_NUMBER, LAG), and aggregations to uncover insights on customer behavior, order patterns, and cohort retention, optimizing queries for large-scale data processing.

WORKSHOPS

Lean for Programmers, Indian Institute of Science (IISc), Bangalore

Dec 2025

- Attended intensive workshop at IISc on applying Lean methodology to software development.

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- Gained hands-on experience in theorem proving, formal verification, and interactive programming in Lean, enhancing skills in rigorous code analysis and proof-assisted development.
 - Explored Lean 4 features for dependently-typed programming and its applications in sustainable software verification.

Formal Methods Update Meeting 2026, Krea University, Sri City

Jul 2 – 3, 2026

- Participated as an attendee in this annual meeting of the Indian Formal Methods community, engaging with presentations and discussions on hardware/software verification, model checking, and theorem proving.